

Ecommerce Sales Trend Analysis Report

1. Introduction to Sales Trend Analysis

Sales trend analysis is a crucial business process that involves examining historical sales data to identify patterns, market shifts, and seasonal variations. By understanding these trends, businesses can predict future performance, optimize inventory levels, and refine marketing strategies. This report analyzes the provided Ecommerce dataset to uncover performance metrics across time, categories, and customer segments.

2. Statistics of the Dataset

The dataset used is an Ecommerce sales prediction sourced from Kaggle.

The dataset contains 1000 transactions spanning from 2023-01-01 to 2025-09-26.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  1000 non-null   object
1   Product_Category      1000 non-null   object
2   Price                 1000 non-null   float64
3   Discount              1000 non-null   float64
4   Customer_Segment      1000 non-null   object
5   Marketing_Spend        1000 non-null   float64
6   Units_Sold            1000 non-null   int64
dtypes: float64(3), int64(1), object(3)
memory usage: 54.8+ KB
```

	Price	Discount	Marketing_Spend	Units_Sold
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	505.241920	24.919750	4912.830240	29.631000
std	289.791248	14.439951	2840.602656	7.267989
min	14.590000	0.010000	100.300000	5.000000
25%	243.342500	12.995000	2413.020000	24.000000
50%	509.310000	24.965000	4837.245000	30.000000
75%	747.592500	37.562500	7319.675000	34.000000
max	999.420000	49.920000	9995.620000	57.000000

```
Index(['Date', 'Product_Category', 'Price', 'Discount', 'Customer_Segment',
      'Marketing_Spend', 'Units_Sold'],
      dtype='object')
```

RangeIndex(start=0, stop=1000, step=1)

Electronics:

2023 - total sales - 1113 total price - 22551 regular
Total sales - 656 total price - 9674 premium
Total sales - 621 total price - 10675 occasional
2024 - total sales - 648 total price - 9555 regular
Total sales - 806 total price - 10346 premium
Total sales - 790 total price - 14875 occasional
2025 - total sales - 647 total price - 9292 regular
Total sales - 453 total price - 7532 premium
Total sales - 476 total price - 8081 occasional

Total sales in regular - 2408 Total price in regular - 41398
Total sales in premium - 1905 Total price in premium - 27552
Total sales in occasional - 1887 Total price in occasional - 33631

Total sales - 6200 Total price - 102581

Fashion:

2023 - total sales - 576 total price - 11353 regular
Total sales - 461 total price - 7407 premium
Total sales - 829 total price - 14395 occasional
2024 - total sales - 788 total price - 11937 regular
Total sales - 793 total price - 14148 premium
Total sales - 657 total price - 10852 occasional
2025 - total sales - 422 total price - 8552 regular
Total sales - 556 total price - 9237 premium
Total sales - 568 total price - 11385 occasional

Total sales in regular - 1786 Total price in regular - 31842
Total sales in premium - 1810 Total price in premium - 30792
Total sales in occasional - 2054 Total price in occasional - 36632

Total sales - 5650 Total price - 99266

Home Decor:

2023 - total sales - 773 total price - 10888 regular
Total sales - 665 total price - 8964 premium
Total sales - 804 total price - 13785 occasional
2024 - total sales - 602 total price - 11754 regular
Total sales - 489 total price - 8042 premium
Total sales - 583 total price - 11955 occasional
2025 - total sales - 523 total price - 7246 regular
Total sales - 607 total price - 9410 premium
Total sales - 597 total price - 11473 occasional

Total sales in regular - 1898 Total price in regular - 29888
Total sales in premium - 1761 Total price in premium - 26416
Total sales in occasional - 1984 Total price in occasional - 37213

Total sales - 5643 Total price - 93517

Sports:

2023 - total sales - 949 total price - 16464 regular
Total sales - 998 total price - 17510 premium
Total sales - 627 total price - 12041 occasional
2024 - total sales - 781 total price - 12147 regular
Total sales - 691 total price - 12673 premium
Total sales - 631 total price - 11517 occasional
2025 - total sales - 536 total price - 8877 regular
Total sales - 389 total price - 6585 premium
Total sales - 523 total price - 10576 occasional

Total sales in regular - 2266 Total price in regular - 37488
Total sales in premium - 2078 Total price in premium - 36768
Total sales in occasional - 1781 Total price in occasional - 34134

Total sales - 6125 Total price - 108390

Toys:

2023 - total sales - 539 total price - 10050 regular
Total sales - 489 total price - 7538 premium
Total sales - 731 total price - 12353 occasional
2024 - total sales - 879 total price - 15128 regular
Total sales - 936 total price - 16626 premium
Total sales - 816 total price - 13552 occasional
2025 - total sales - 446 total price - 6183 regular
Total sales - 578 total price - 9474 premium
Total sales - 589 total price - 10581 occasional

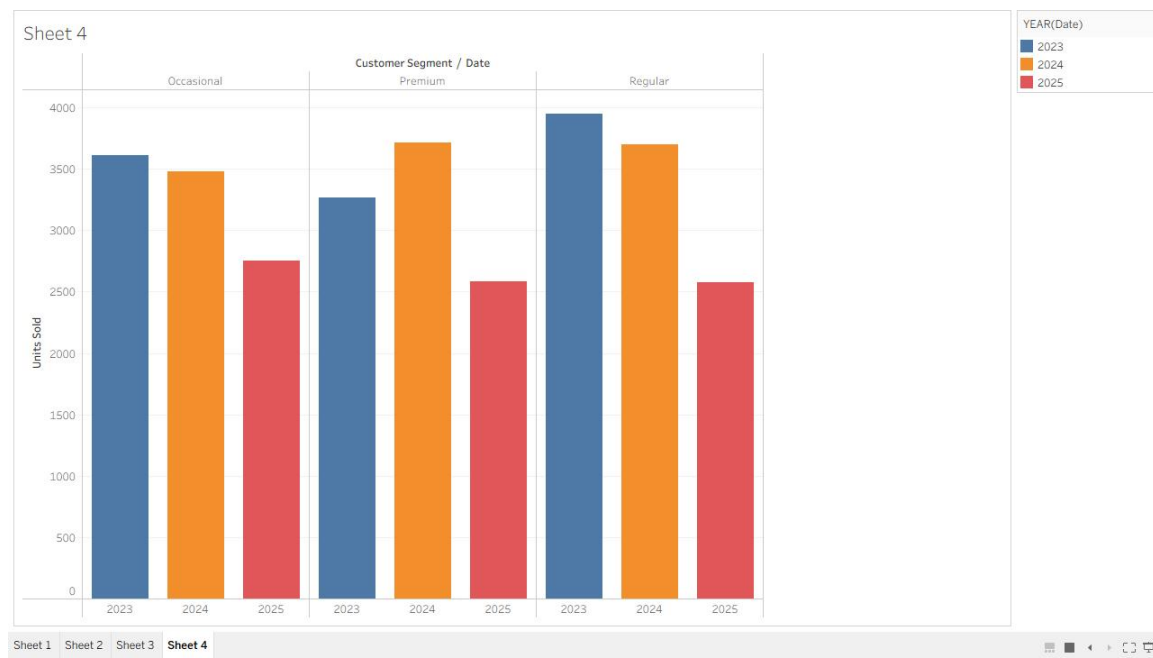
Total sales in regular - 1864 Total price in regular - 31361
Total sales in premium - 2003 Total price in premium - 33638
Total sales in occasional - 2136 Total price in occasional - 36486

Total sales - 6003 Total price - 101485

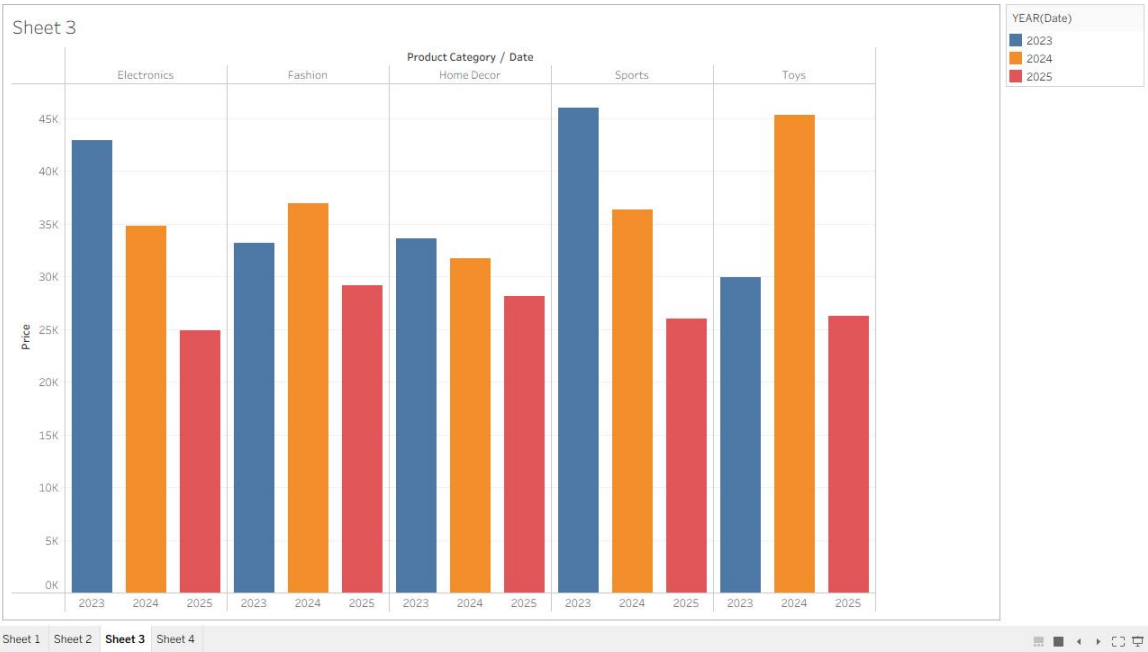
3. Sales Trends and Visualizations

The following charts illustrates the monthly sales performance during the recorded period:

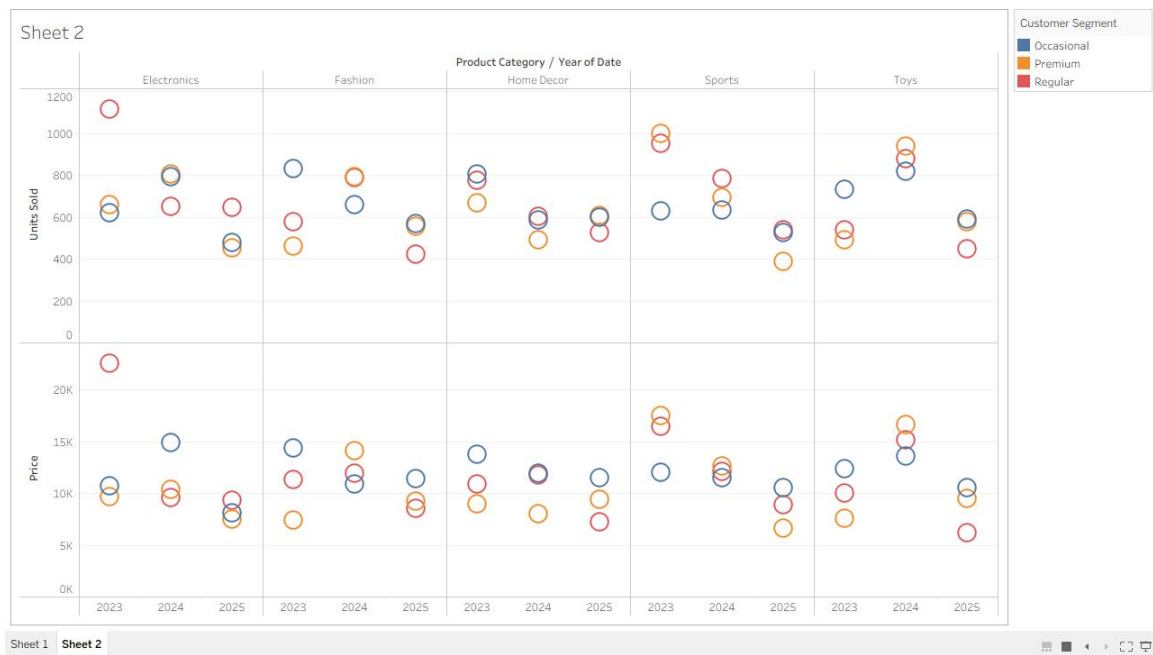
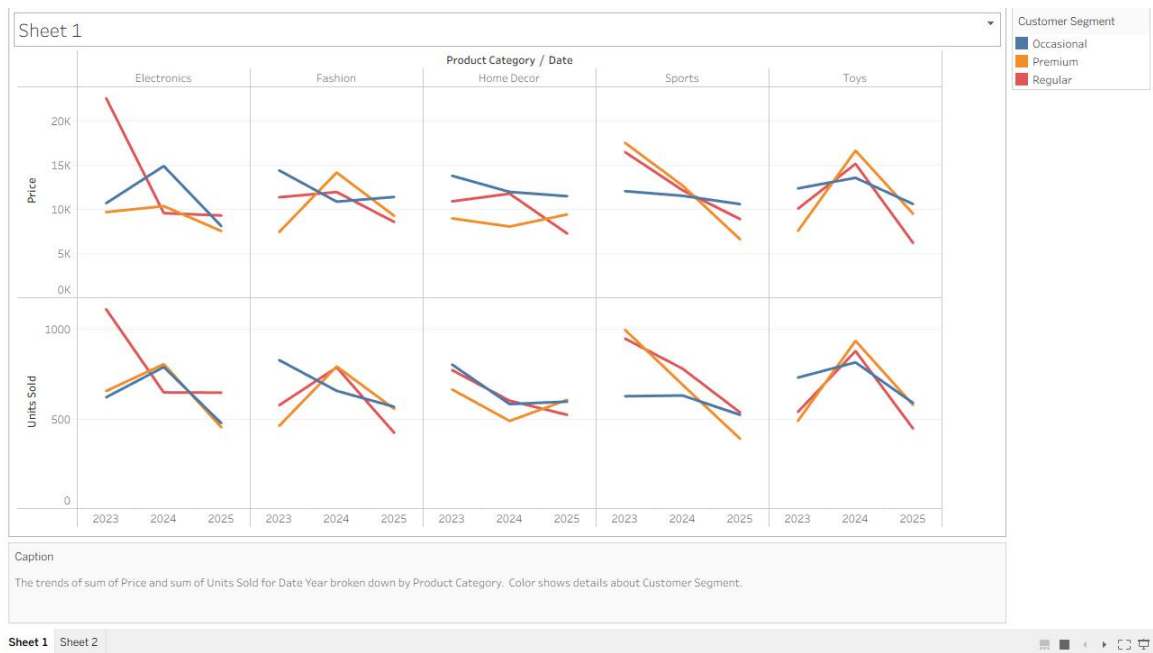
Distribution of sales across different customer segment:



Distribution of sales across different product category:



Distribution of sales across different product categories and different segments and how many units are sold.



4. Key Insights

- Dominant Category: The 'Sports' category is the leading contributor to total revenue.
- High-Value Segment: The 'Regular' customer segment shows the highest engagement/sales volume.
- Marketing Correlation: There is an observable relationship between marketing spend and units sold, suggesting that targeted campaigns drive volume.

5. Recommended Actions

- Inventory Optimization: Increase stock levels for 'Sports' ahead of peak periods identified in the trend chart.
- Loyalty Programs: Develop exclusive offers for the 'Occasional' segment to maximize lifetime value.
- Discount Strategy: Use deep discounts strategically during low-sales months to clear inventory and maintain cash flow.
- Marketing Reallocation: Reallocate budget from underperforming categories to those with higher Return on Investment shown in the analysis.